

C++ Programming Fundamentals

Standard Notes & Quiz Review

About These Notes

This document provides a clean, minimalistic overview of fundamental C++ concepts including memory allocation, compilation mechanisms, object-oriented programming paradigms, and standard library usage. Each assessment module includes a detailed explanation formatted for clarity and retention.

Self-Assessment

Question 1

Which type of memory is used for dynamic memory allocation in C++?

- A. Stack
- B. Heap
- C. Cache
- D. ROM

Correct Answer: B

Explanation: Dynamic memory allocation in C++ is done on the heap using new and delete.

Question 2

_____ translates the entire source code in a single run. It consumes less time i.e., it is faster than _____.

- A. Compiler, Linker
- B. Interpreter, Compiler
- C. Linker, Compiler
- D. Compiler, Interpreter

Correct Answer: D

Explanation: A compiler translates the entire source code in a single run, making it faster. An interpreter translates the program line by line, which makes it slower compared to a compiler.

Question 3

Multi-line comments in C++ are defined as

- A. `//`
- B. `/** */`
- C. `/* */`
- D. Both (A) and (B)

Correct Answer: C

Explanation: Single-line comments start with `//` and extend to the end of the line. Multi-line comments start with `/*` and end with `*/`.

Question 4

Why is C++ faster than Java?

- A. Because C++ takes fewer lines of code as compared to Java.
- B. C++ is a compiled language whereas Java is an interpreted language.
- C. C++ is closer to the hardware.
- D. Both (B) and (C)

Correct Answer: D

Explanation: C++ is generally considered to be faster than Java because it is a compiled language, while Java is an interpreted language. When a program written in a compiled language such as C++ is executed, the source code is first transformed into machine code by a compiler. This machine code is a set of low-level instructions that can be directly executed by the CPU. Because the machine code is specific to the computer's architecture, it can be executed very efficiently. On the other hand, when a program written in an interpreted language such as Java is executed, it is not compiled into machine code. Instead, it is interpreted by a program called the Java Virtual Machine (JVM) as it is being executed. The JVM reads the source code and translates it into machine code on the fly, which is then executed by the CPU. Because the JVM has to interpret the source code at runtime, this can be slower than executing pre-compiled machine code. As a result, Java programs may not run as quickly as C++ programs. Also, C++ is closer to the hardware.

Question 5

Which of the following describes object-oriented programming in C++?

- A. Dividing code into multiple files
- B. Using objects to represent real-world entities
- C. Writing programs using only functions
- D. Creating software without data

Correct Answer: B

Explanation: Object-Oriented Programming (OOP) uses objects to model real-world entities and their interactions. This approach organizes data and methods into classes and objects.

Question 6

Which of the following is a better practice than using namespace std; in C++?

- A. Using global variables
- B. Writing custom namespaces
- C. Using std:: explicitly before standard library objects
- D. Removing all namespaces

Correct Answer: C

Explanation: While using namespace std; is convenient, it can lead to name collisions in large programs. Using std::cout or std::cin explicitly avoids such issues and is considered better practice.

Question 7

Why are comments used in C++?

- A. To make the code beautiful.
- B. Improve readability
- C. Provide documentation
- D. Both (b) and (c)

Correct Answer: D

Explanation: Comments are used in C++ to provide explanations and annotations in the source code. They are ignored by the compiler and do not have any effect on the behavior of the program.

Question 8

Predefined words in a programming language that have a special meaning and cannot be used as a variable name or function name are called

- A. data types
- B. keywords
- C. predefined words
- D. literals

Correct Answer: B

Explanation: Keywords are predefined words in a programming language that have a special meaning and cannot be used as a variable name or function name. eg, "int", "float", "const", etc.

Question 9

Which of the following best describes C++ as a programming language?

- A. High-level language Only
- B. Low-level language
- C. Middle-level language
- D. None of these

Correct Answer: C

Explanation: C++ is often referred to as a middle-level language because it supports both high-level features (like object-oriented programming) and low-level features (like direct memory access), allowing fine control over system resources.

Question 10

Which header file must be included to use `std::cin` and `std::cout`?

- A. `<math.h>`
- B. `<input>` `<output>`
- C. `<iostream>`
- D. `<program>`

Correct Answer: C

Explanation: The `<iostream>` header file defines input and output stream objects like `std::cin` and `std::cout`, which are used for standard input and output operations in C++.

Question 11

Which of the following is a valid C++ keyword?

- A. `define`
- B. `include`
- C. `class`
- D. `macro`

Correct Answer: C

Explanation: class is a C++ keyword used to define a class. The others are either preprocessor directives or not valid C++ keywords.

Question 12

What is the correct way to declare a constant variable in C++?

- A. `const int x = 5;`
- B. `int const = 5;`
- C. `constant int x = 5;`
- D. `define int x = 5;`

Correct Answer: A

Explanation: `const` keyword should be used with the declaration of a variable to declare a constant variable in C++.

Question 13

Which of the following is not a C++ feature?

- A. Dynamic Memory Access.
- B. Object Oriented Programming
- C. Garbage Collection
- D. Operator Overloading

Correct Answer: C

Explanation: C++ doesn't have built-in garbage collection like Java, In C++ the memory management is mostly manual.

Question 14

What is the correct file extension for C++ source files?

- A. .c
- B. .cpp
- C. .h
- D. .txt

Correct Answer: B

Explanation: .cpp is the file extension used for C++ source code files.

Question 15

Which of the following is a C++ keyword used to define a user-defined type?

- A. function
- B. define
- C. class
- D. typedef

Correct Answer: C

Explanation: class is a keyword in C++ used to define user-defined types (i.e. classes).

Question 16

What is the default access specifier for class members in C++?

- A. public
- B. private
- C. protected
- D. global

Correct Answer: B

Explanation: In C++, class members are private by default.

Question 17

What is the use of header file `<iostream>` in C++?

- A. Declares functions
- B. Enable Adding comments
- C. Enables input/output operations
- D. Allocating memory

Correct Answer: C

Explanation: The `<iostream>` header file contains declarations for `cin`, `cout`, etc which enable input-output operations. .

Question 18

What is the size of an empty class in C++?

- A. 0 bytes
- B. 1 bytes
- C. 2 bytes
- D. Compiler Dependent

Correct Answer: B

Explanation: In C++ even empty classes occupy at least 1 byte to ensure the uniqueness of object addresses.

Question 19

Which of the following correctly differentiates C and C++?

- A. C++ is a procedural language, while C is object-oriented
- B. C supports classes and objects, C++ does not
- C. C++ supports both procedural and object-oriented programming, while C supports only procedural programming
- D. C has features like inheritance and polymorphism, C++ does not

Correct Answer: C

Explanation: C++ supports both procedural and object-oriented programming (including features like inheritance, polymorphism, and encapsulation), whereas C supports only procedural programming.

Question 20

What will this code print?

```
#include <iostream>
using namespace std;
int main() {
    int x = 5;
    cout << x;
    return 0;
}
```

- A. 5
- B. 6
- C. Error
- D. Undefined

Correct Answer: A

Explanation: The above program declares an integer type variable and prints it on the console.